TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Fabric Strength (Tensile, Tearing, Ballistic and Bursting)

One fabric, four ways to break it: the ravelled strip pulled, the Elmendorf pendulum tearing, the diaphragm bursting — and why each failure mode needs its own test.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 6 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the strength of the reference fabric by three standard procedures — the ravelled-strip tensile test (ISO 13934-1), the Elmendorf ballistic tearing test (ISO 13937-1) and the hydraulic bursting test with diaphragm correction (ISO 13938) — and to interpret the four strength values (warp tensile, weft tensile, tearing, bursting) as one profile.

1.1 Why four strengths?

Fabric in service is rarely pulled evenly in two hands. Garments fail by steady tension at seams and stress points (tensile), by a snag that concentrates force onto one or two threads and runs (tearing), and by biaxial bulging at knees, elbows and upholstery (bursting). Each mode loads a different thread geometry, so no single number predicts the others. The **ballistic** principle — a falling pendulum whose lost energy does the tearing — belongs to the tearing test: the pendulum's energy loss measures the work of tearing, which is why the Elmendorf instrument is classed with ballistic methods.

1.2 Tensile strength — ravelled strip (ISO 13934-1)

In the strip method a specimen is ravelled down to an exact working width of 50 mm (threads removed from both edges of a wider cutting so the counted threads are undamaged), clamped over a 200 mm gauge length and extended at constant rate (100 mm/min) until rupture. The maximum force is the breaking load. Ravelled strip is preferred over cut strip for woven fabrics because cut edges either include partial threads or lose edge threads during the test. Warp-way and weft-way specimens are tested separately: the warp, more densely set (70 EPI against 59 PPI here) and held under weaving tension history, normally breaks higher.

$$\text{Strength per cm} = \text{breaking load of 50 mm strip (N)} / 5\ \text{cm}$$

1.3 Tearing strength — Elmendorf (ISO 13937-1)

A specimen is slit part-way, clamped half in a fixed jaw and half in a pendulum jaw; the raised pendulum is released, swings down and tears the specimen through the remaining distance. The energy the pendulum loses — read from its scale, calibrated in newtons — is the tearing force. Tearing strength is a *one-thread-at-a-time* property: at the tear tip only a small group of threads (the del) carries the load, so yarn mobility, weave float and thread grouping matter as much as yarn strength. A fabric can be tensile-strong and tear-weak if its threads cannot bunch together to share the tear load.

1.4 Bursting strength (ISO 13938)

The specimen is clamped over a circular rubber diaphragm; hydraulic pressure beneath the diaphragm distends both until the fabric bursts. The gauge reads total pressure at burst, part of which was spent distending the diaphragm alone. A diaphragm correction — the pressure needed to distend the bare diaphragm to the same height — is subtracted:

$$\text{Bursting strength} = \text{total pressure at burst} - \text{diaphragm pressure}$$

Bursting is direction-free: warp and weft are loaded together biaxially, which makes it the natural strength test for knits and the fairest single comparison for fabrics of unknown weak direction.

All specimens are conditioned and tested at 65% RH and 20 °C.

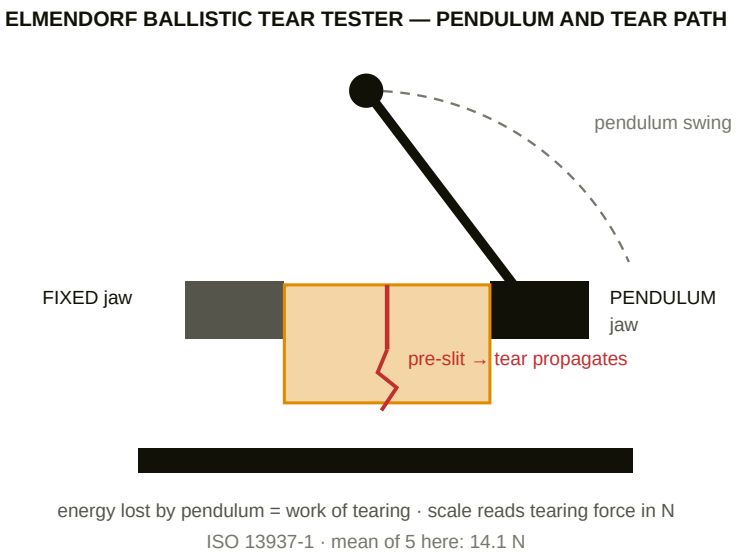


Figure 1 — The Elmendorf ballistic pendulum tear tester: the specimen bridges the fixed jaw and the pendulum jaw across a pre-cut slit; the released pendulum tears it through, and the pendulum's energy loss, read on the calibrated scale, gives the tearing force.

2. Apparatus and procedure

- **Apparatus:** constant-rate-of-extension tensile tester with 200 mm gauge (ISO 13934-1); Elmendorf tearing tester (ISO 13937-1) with calibrated pendulum and specimen cutter; hydraulic bursting tester with diaphragm (ISO 13938); specimen templates, raveling needle; conditioned reference fabric (70 × 59, 185.5 g/m²).
- **Strip tensile:** cut specimens oversize, ravel both edges to exactly 50 mm width, mount straight and untwisted at 200 mm gauge, extend at 100 mm/min to rupture and record maximum force. Test five warp-way and five weft-way specimens; reject breaks within 5 mm of a jaw.
- **Elmendorf tear:** cut specimens to the template, make the standard initial slit, clamp one half in the fixed jaw and one in the pendulum jaw, raise and release the pendulum, and read the tearing force from the scale. Test five specimens in the direction reported and check that each tear followed the slit line.
- **Bursting:** clamp the specimen over the diaphragm without pre-distension, raise hydraulic pressure at the standard rate until burst, and record total pressure. Without fabric, distend the bare diaphragm to the same mean height and record the diaphragm pressure. Subtract for bursting strength.
- **Reporting:** average each set, express strip results also per centimetre of width, and present all four values together as the fabric's strength profile.

3. Worked calculation

Strip tensile (ISO 13934-1). Warp-way breaks (N): 412, 428, 405, 435, 420.

$$\text{Mean warp breaking load} = (412 + 428 + 405 + 435 + 420) / 5 = 2{,}100 / 5 = 420 \text{ N per 50 mm} = 84 \text{ N/cm}$$

Weft-way breaks (N): 318, 330, 322, 336, 314.

$$\text{Mean weft breaking load} = (318 + 330 + 322 + 336 + 314) / 5 = 1{,}620 / 5 = 324 \text{ N per 50 mm} = 64.8 \text{ N/cm}$$

The warp/weft ratio, 420/324 ≈ 1.30, sits close to the density ratio 70/59 ≈ 1.19 with the remainder from the warp's stronger weaving history and slightly coarser yarn (Ne 20 against Ne 24, Experiment 5) — construction explains the profile.

Elmendorf tear (ISO 13937-1). Readings (N): 14.6, 13.9, 14.2, 13.5, 14.3.

$$\text{Mean tearing strength} = (14.6 + 13.9 + 14.2 + 13.5 + 14.3) / 5 = 70.5 / 5 = 14.1 \text{ N}$$

Bursting (ISO 13938).

$$\text{Bursting strength} = 320 \text{ kPa (total)} - 28 \text{ kPa (diaphragm)} = 292 \text{ kPa}$$

TEST	STANDARD	MEAN RESULT
Tensile, warp-way (50 mm strip)	ISO 13934-1	420 N (84 N/cm)
Tensile, weft-way (50 mm strip)	ISO 13934-1	324 N (64.8 N/cm)
Tearing (Elmendorf)	ISO 13937-1	14.1 N
Bursting (corrected)	ISO 13938	292 kPa

ONE PROFILE, FOUR NUMBERS

Tensile 420/324 N, tear 14.1 N, bursting 292 kPa describe a sound medium-weight apparel fabric, warp-dominant exactly as its 70 × 59 construction predicts. A specification failure in only one row would point the investigation: tensile low in both directions suggests weak yarn; tear low alone suggests a tight, immobile weave; bursting low with tensile normal suggests a localised weak direction or damage.

4. Precautions

- Ravel strip specimens to the exact 50 mm of intact threads; a half-thread error at each edge is a 2% load error with no visible sign.
- Reject jaw breaks and slit-deviating tears and replace them — both measure the clamping or the cutting, not the fabric.
- Check the Elmendorf pendulum capacity: the mean reading should fall in the middle working range of the scale; add or remove pendulum weights to achieve this rather than reading at the scale's extreme ends.
- Distend the bare diaphragm to the same height as the burst for the correction; a correction taken at a different distension misstates the 28 kPa subtraction.
- Mount strip specimens dead straight; an angled specimen loads its edge threads first and fails progressively at a low, ragged peak.
- Keep the standard atmosphere (65% RH, 20 °C) throughout: cotton strength rises with regain, so an unconditioned set is not comparable with the conditioned specification.

Practice (viva preparation)

1. Find the mean warp and weft strip breaking loads from: warp 412, 428, 405, 435, 420 N; weft 318, 330, 322, 336, 314 N. Express the warp value per cm.
2. Why is the warp-way strength higher than the weft-way strength in this fabric?
3. Find the mean Elmendorf tearing strength from 14.6, 13.9, 14.2, 13.5 and 14.3 N.

4. Total bursting pressure is 320 kPa and the diaphragm correction is 28 kPa. Find the bursting strength and explain the correction.
5. Why is a ravelled strip preferred to a simply cut strip for woven fabric tensile testing?
6. A fabric is tensile-strong but tear-weak. Explain how that combination arises.
7. Why is bursting strength the natural test for knitted fabrics?

Answer key

1. Warp: 2,100 / 5 = 420 N per 50 mm strip = 420 / 5 = 84 N/cm. Weft: 1,620 / 5 = 324 N per 50 mm strip (64.8 N/cm).
2. Three construction reasons act together: the warp is more densely set (70 EPI against 59 PPI — more threads share each strip's load); the warp yarn is slightly coarser (Ne 20 against Ne 24); and warp yarns are sized and woven under tension, selected and prepared for strength in a way weft yarns are not. The measured ratio of about 1.30 matches this construction.
3. Mean = 70.5 / 5 = 14.1 N.
4. Bursting strength = 320 − 28 = 292 kPa. Part of the total pressure was consumed in distending the rubber diaphragm itself to the burst height; only the remainder distended and burst the fabric. The bare-diaphragm pressure at the same distension is therefore measured separately and subtracted (ISO 13938).
5. A cut strip either includes damaged partial threads at its edges or loses edge threads that fray out during extension, so the tested width is not the nominal width. Ravelling removes threads from a wider cutting until exactly 50 mm of whole, undamaged threads remains — the width is defined by counted threads, not by the scissors.
6. Tensile loading shares force across all threads in the strip width; tearing concentrates force at the tear tip onto the small thread group of the del. If the weave is tight and the yarns cannot slide and bunch into the del — high cover, long size film, resin finish — few threads carry the tear and each snaps in turn at low force, while the same fabric's tensile value, using every thread, stays high. Strength of yarn and mobility of structure are separate variables.
7. A knit has no warp/weft pair of straight thread systems to ravel into strips, and a cut strip of knit curls, ladders and extends enormously before breaking, giving a geometry-dependent result. Bursting loads the fabric biaxially in its own plane — the way knits are loaded on the body — and gives one direction-free strength value, which is why ISO 13938 is the standard strength route for knitted fabrics.
